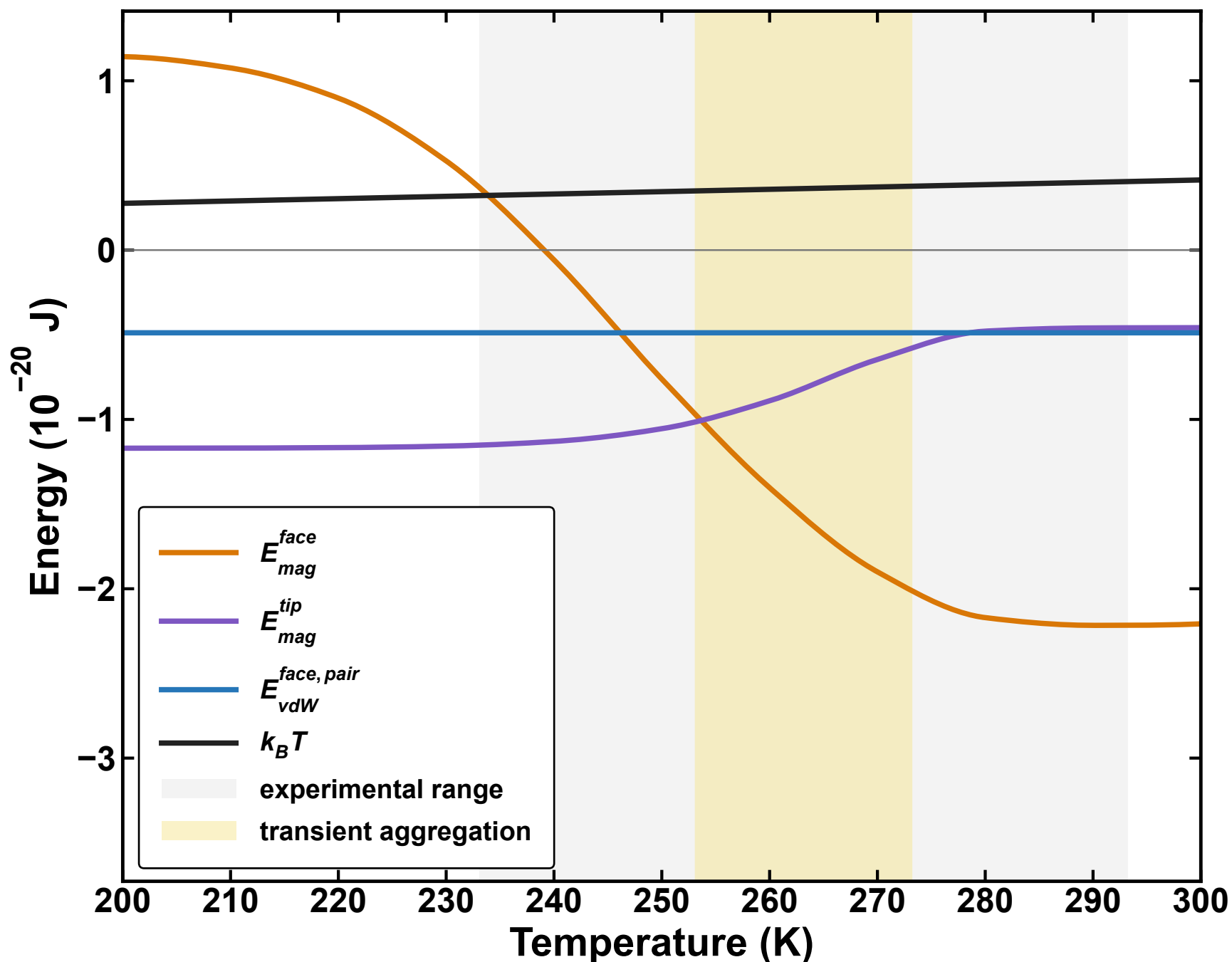


Pair energies with rigid-cube Brownian rotation 16 nm Fe₃O₄, 1.5 nm ligand shell; 20 s window



Each temperature starts from a random 64-state prior; blocked $\langle 111 \rangle$ moments may align by cube rotation.
The face curve includes the Néel relaxation deficit and suppressed-rotation penalty; no barrier is fitted.